

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 1: Rational and Irrational Numbers

Topic-wise short notes with solved examples

Rational numbers and their decimal forms, irrational numbers and surds, placing them on the number line, simplifying surds and rationalising denominators.

## 1 Rational numbers

- A **rational number** can be written as  $\frac{p}{q}$ , where  $p$  and  $q$  are integers and  $q \neq 0$ .
- Every integer, every fraction, every terminating decimal and every recurring decimal is rational.
- Between any two rational numbers there are infinitely many rational numbers. One of them is their mean:  $\frac{a+b}{2}$ .
- To insert  $n$  rational numbers between  $a$  and  $b$ , write both with a common denominator, multiply numerator and denominator by  $n+1$ , and pick the fractions in between.

**Example 1.1** Insert three rational numbers between  $\frac{1}{3}$  and  $\frac{1}{2}$ .

SOLUTION

$$\frac{1}{3} = \frac{2}{6} = \frac{8}{24} \text{ and } \frac{1}{2} = \frac{3}{6} = \frac{12}{24}$$

$$\frac{9}{24}, \frac{10}{24}, \frac{11}{24}, \text{ that is } \frac{3}{8}, \frac{5}{12}, \frac{11}{24}$$

## 2 Decimal form of a rational number

- The decimal form of a rational number either **terminates** or **recurs**.
- $\frac{p}{q}$  (in lowest terms) terminates exactly when  $q$  has **no prime factor other than 2 and 5**.
- A recurring decimal is written with a bar:  $0.\overline{3} = 0.333\dots$ ,  $0.1\overline{6} = 0.1666\dots$
- Recurring decimal to fraction:** let  $x$  be the number, multiply by a power of 10 so that the recurring parts line up, and subtract.

**Example 2.1** Without dividing, say whether  $\frac{7}{80}$  and  $\frac{5}{12}$  terminate.

SOLUTION

$$80 = 2^4 \times 5; \text{ only 2 and 5. } 12 = 2^2 \times 3; \text{ contains 3.}$$

$$\frac{7}{80} \text{ terminates; } \frac{5}{12} \text{ recurs}$$

**Example 2.2** Express  $0.\overline{27}$  as a fraction.

SOLUTION

Let  $x = 0.2727 \dots$ . Then  $100x = 27.2727 \dots$

$$99x = 27$$

$$x = \frac{27}{99} = \frac{3}{11}$$

**Example 2.3** Express  $0.2\overline{3}$  as a fraction.

SOLUTION

$10x = 2.333 \dots$  and  $100x = 23.333 \dots$ , so  $90x = 21$

$$\frac{7}{30}$$

### 3 Irrational numbers

- A number that **cannot** be written as  $\frac{p}{q}$  is **irrational**. Its decimal form neither terminates nor recurs.
- Examples:  $\sqrt{2}$ ,  $\sqrt{3}$ ,  $\sqrt{5}$ ,  $\sqrt[3]{4}$ ,  $\pi$ ,  $0.101001000100001 \dots$
- $\sqrt{n}$  is irrational whenever  $n$  is a positive integer that is not a perfect square.
- Rational and irrational numbers together form the **real numbers**. Every real number has a point on the number line.
- **Proof that  $\sqrt{2}$  is irrational:** suppose  $\sqrt{2} = \frac{p}{q}$  in lowest terms. Then  $p^2 = 2q^2$ , so  $p$  is even, say  $p = 2k$ . Then  $q^2 = 2k^2$ , so  $q$  is even too. This contradicts "lowest terms".

| Operation                             | Result                        |
|---------------------------------------|-------------------------------|
| rational $\pm$ irrational             | irrational                    |
| non-zero rational $\times$ irrational | irrational                    |
| irrational $\pm$ irrational           | may be rational or irrational |
| irrational $\times$ irrational        | may be rational or irrational |

**Example 3.1** Show that  $3 + \sqrt{5}$  is irrational.

SOLUTION

Suppose  $3 + \sqrt{5} = r$ , a rational number. Then  $\sqrt{5} = r - 3$ , which is rational.

**This contradicts the fact that  $\sqrt{5}$  is irrational, so  $3 + \sqrt{5}$  is irrational**

**Example 3.2** Write two irrational numbers between 2 and 3.

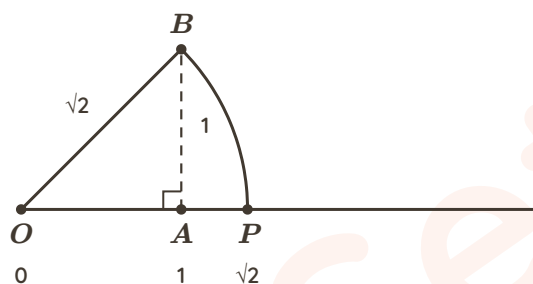
SOLUTION

Any  $\sqrt{n}$  with  $4 < n < 9$  and  $n$  not a perfect square will do.

$\sqrt{5}$  and  $\sqrt{7}$

#### 4 Surds on the number line

- $\sqrt{2}$ : draw a right triangle with both shorter sides 1 unit on the number line from 0. Its hypotenuse is  $\sqrt{2}$ . With centre 0 and this radius, cut the number line.
- $\sqrt{3}$ : use shorter sides  $\sqrt{2}$  and 1.  $\sqrt{5}$ : use shorter sides 2 and 1.



OA = 1 and AB = 1, so OB =  $\sqrt{2}$ . The arc with centre O marks P at  $\sqrt{2}$ .

#### 5 Surds

- If  $\sqrt[n]{a}$  is irrational and  $a$  is a positive rational number, then  $\sqrt[n]{a}$  is a **surd** of order  $n$ .  $\sqrt{3}$  is a surd;  $\sqrt{9}$  and  $\pi$  are not.
- $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$  and  $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$  for positive  $a, b$ .
- **Like surds** have the same irrational part and can be added:  $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$ .
- **Simplest form**: take out the largest perfect-square factor.  $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$ .

**Example 5.1** Simplify  $\sqrt{50} + \sqrt{18} - \sqrt{8}$ .

SOLUTION

$$5\sqrt{2} + 3\sqrt{2} - 2\sqrt{2}$$

$$6\sqrt{2}$$

**Example 5.2** Expand  $(2\sqrt{3} + \sqrt{5})(2\sqrt{3} - \sqrt{5})$ .

SOLUTION

$$(2\sqrt{3})^2 - (\sqrt{5})^2 = 12 - 5$$

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**Example 5.3** Which is greater,  $3\sqrt{2}$  or  $2\sqrt{5}$ ?

SOLUTION

$$3\sqrt{2} = \sqrt{18} \text{ and } 2\sqrt{5} = \sqrt{20}$$

$$2\sqrt{5}$$

## 6 Rationalisation

- **Rationalising** a denominator means removing the surd from it by multiplying numerator and denominator by a suitable **rationalising factor**.

| Denominator           | Rationalising factor  |
|-----------------------|-----------------------|
| $\sqrt{a}$            | $\sqrt{a}$            |
| $a + \sqrt{b}$        | $a - \sqrt{b}$        |
| $\sqrt{a} + \sqrt{b}$ | $\sqrt{a} - \sqrt{b}$ |
| $\sqrt{a} - \sqrt{b}$ | $\sqrt{a} + \sqrt{b}$ |

- $a + \sqrt{b}$  and  $a - \sqrt{b}$  are **conjugates**. Their product  $a^2 - b$  is rational.

**Example 6.1** Rationalise the denominator of  $\frac{5}{\sqrt{3}}$ .

SOLUTION

$$\frac{5\sqrt{3}}{3}$$

**Example 6.2** Rationalise the denominator of  $\frac{4}{3 - \sqrt{5}}$ .

SOLUTION

$$\frac{4(3 + \sqrt{5})}{9 - 5}$$

$$3 + \sqrt{5}$$

**Example 6.3** If  $\frac{\sqrt{3} + 1}{\sqrt{3} - 1} = a + b\sqrt{3}$ , find  $a$  and  $b$ .

SOLUTION

$$\frac{(\sqrt{3} + 1)^2}{3 - 1} = \frac{4 + 2\sqrt{3}}{2} = 2 + \sqrt{3}$$

$$a = 2, b = 1$$

**Example 6.4** If  $x = 2 + \sqrt{3}$ , find  $x + \frac{1}{x}$  and  $x^2 + \frac{1}{x^2}$ .

SOLUTION

$$\frac{1}{x} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3}, \text{ so } x + \frac{1}{x} = 4.$$

$$x^2 + \frac{1}{x^2} = 4^2 - 2$$

**4 and 14**

## 7 Common mistakes

- Writing  $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ . This is false:  $\sqrt{9+16} = 5$ , not 7.
- Calling  $\frac{22}{7}$  irrational. It is rational; it is only an approximation for  $\pi$ .
- Adding unlike surds:  $\sqrt{2} + \sqrt{3}$  is not  $\sqrt{5}$ .
- Multiplying only the denominator by the rationalising factor.
- Using  $a + \sqrt{b}$  again instead of its conjugate.
- Forgetting to reduce  $\frac{p}{q}$  to lowest terms before testing the denominator for 2s and 5s.